What Drives First-Year MLB Arbitration Salaries for Starting Pitchers?

A Fangraphs + Spotrac + Statcast Econometric Analysis (2016-2026)

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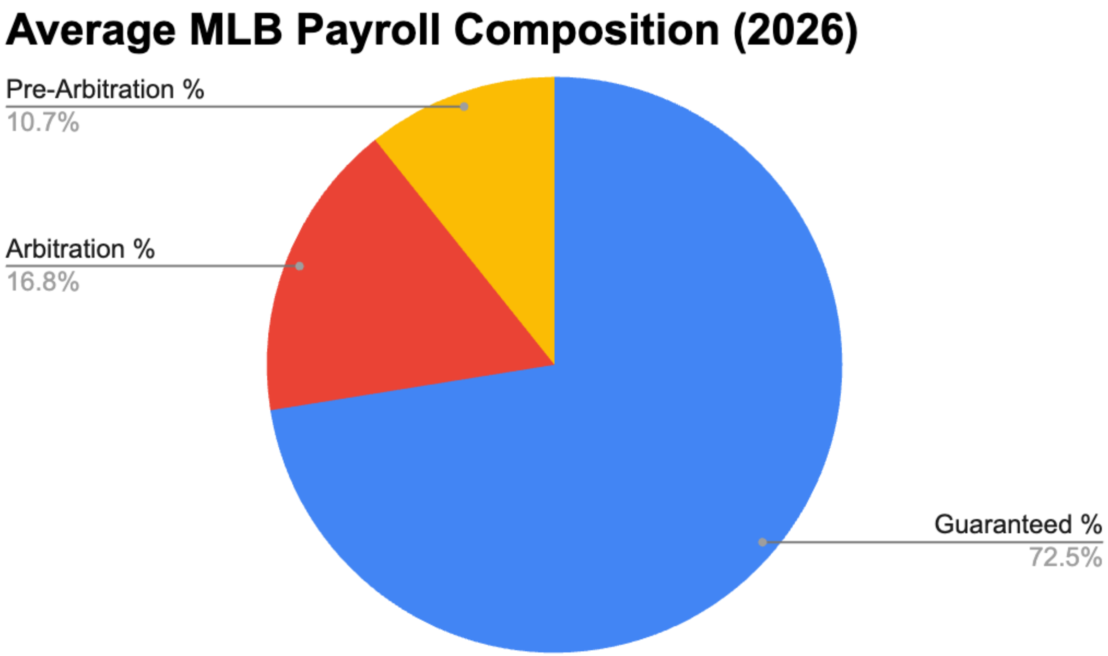
Why This Matters?

- MLB arbitration affects millions in payroll decisions. Teams and players argue using performance metrics, but which metrics influence first-year salaries?



Data Obtained via Fangraphs Payroll Breakdown, Author Calculations

Data

Component	Source
Arbitration Salaries	Spotrac
Pitching Performance	Statcast
Sample	Arbitration 1 + Super Two SP
Years	2016-2026
Observations	124

Source: Spotrac, Fangraphs, Statcast, Author calculations

Descriptive Statistics

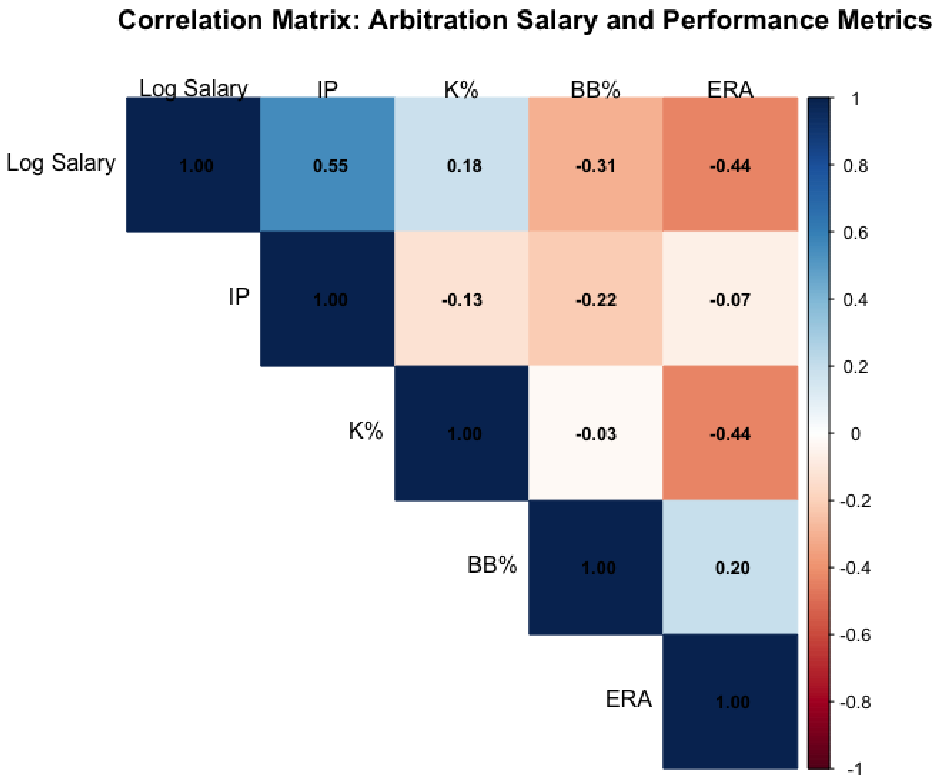
Variable	Mean	SD	Min	Max
Salary	$2.89M	$1.09M	$0.88M	$7.25M
IP	132	43	51	232
K%	22.8	4.81	12.6	35.6
BB%	7.6	1.8	3	13.5
ERA	3.96	0.97	1.78	6.73

- Salary distributions are heavily right-skewed, motivating a logarithmic specification

Source: Spotrac, Fangraphs, Statcast, Author calculations

Correlation Heatmap

- Innings Pitched and Log Salary exhibit the strongest positive relationship



Source: Spotrac, Fangraphs, Statcast, Author calculations

Model Specification

$$\log(Salary_i) = \beta_0 + \beta_1 (IP / 10)_i + \beta_2 K_i + \beta_3 BB_i + \beta_4 ERA_i + \delta_t + u_i$$

- Salary modeled in logs
- Year fixed effects control for differences in salary environments across arbitration classes (2016-2026)
- Prior-year performance predicts arbitration outcomes

δt = Year fixed effects
u_i = *error term*

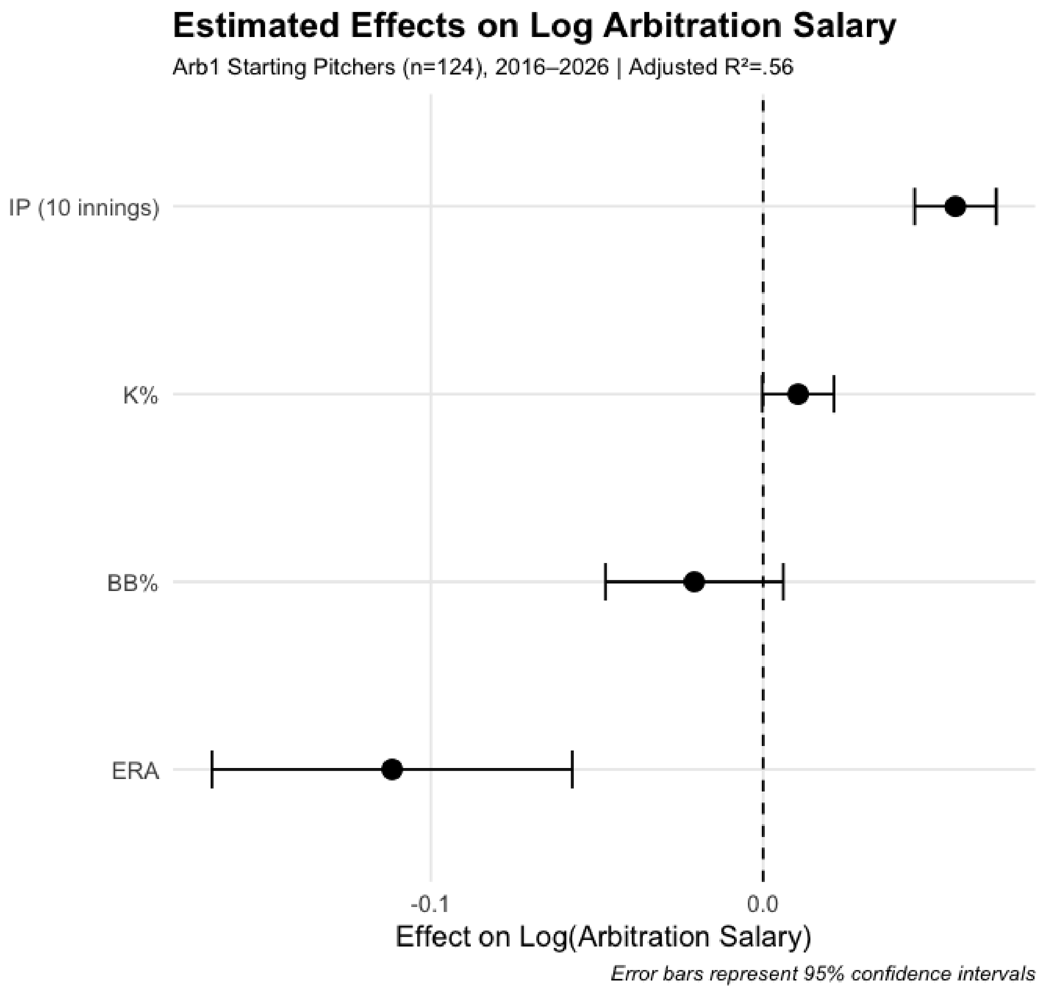
Results

Adjusted $R^2 = .56$
N = 124

Variable	Effect	p-value
IP (10 Innings)	+5.8%	< .001
K%	+1.05%	.055
BB%	-2.1%	.127
ERA	-11.2%	< .001

Source: Spotrac, Fangraphs, Statcast, Author calculations

Coefficient Plot



Source: Spotrac, Fangraphs, Statcast, Author calculations

Baseball Interpretation

- Key Findings

  Estimated average effects holding performance and season constant
- Workload dominates
  - 10 additional innings pitched → ~ 5.8% higher salary
- ERA still matters
  - One-run ERA increase → ~ 11% lower salary
- Strikeouts help
  - Moderate evidence
- Walks appear less influential
- Arbitration may reward traditional workload and run-prevention metrics more heavily than advanced indicators

Year fixed effects included; error bars represent 95% confidence intervals

Model Successes and Failures

Largest Underpredictions

Pitcher	Season	Predicted	Actual	Difference
Shane Bieber	2022	$2.65M	$6.0M	+$3.35M
Corbin Burnes	2022	$4.74M	$6.5M	+$1.76M
Hunter Brown	2023	$4.35M	$5.71M	+$1.36M

- Interpretation
  - The largest underpredictions occur among elite pitchers, suggesting arbitration awards may incorporate factors not fully captured by traditional performance metrics

Source: Spotrac, Fangraphs, Statcast, Author calculations

Model Successes and Failures

Largest Overpredictions

Pitcher	Season	Predicted	Actual	Difference
Michael King	2023	$2.09M	$1.3M	-$788K
Cal Quantrill	2022	$3.26M	$2.51	-$752K

- Interpretation
  - Performance metrics alone may not capture role or context

Source: Spotrac, Fangraphs, Statcast, Author calculations

Limitations

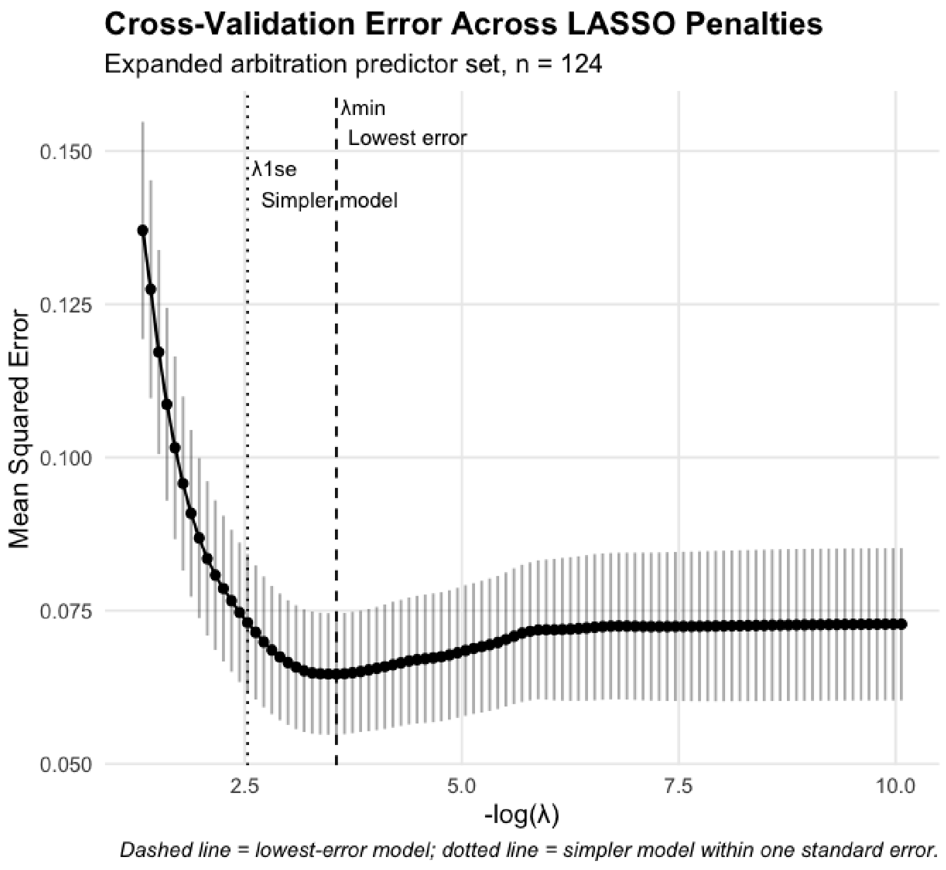
- Possible omitted variables
  - WAR
  - Awards
  - Injuries
  - Reputation
  - Role
  - Comparable-player effects
  - Advanced Statcast metrics excluded from baseline specification
- Violations of:
  $E(u \mid X) = 0$ → may create omitted variable bias

Conclusion

- Workload and run prevention significantly influence first-year arbitration salaries
- Arbitration appears to reward traditional performance outcomes, although advanced contact-quality metrics also demonstrate predictive value
- Elite pitchers may receive premiums beyond measurable performance characteristics

LASSO Extension

- Takeaways
  - Cross-Validation used to select the penalty parameter (λ)
  - Simplest model within one standard error retained:
    - Quality Starts
    - xwOBA
    - Barrel Rate
  - RMSE = .239 (vs. .231 for OLS)
  - LASSO emphasized workload outcomes and contact-quality measures while shrinking traditional rate statistics toward zero



Source: Spotrac, Fangraphs, Statcast, Author calculations